OmniCart

An Intelligent E-Commerce Recommender System with LLM Integration and Automated Workflows

AI COURSE — PROJECT PROPOSAL

Project Title	***OmniCart***: An Intelligent E-Commerce Recommender System with LLM Integration and Automated Workflows
Course	Artificial Intelligence
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1. Project Overview

1.1 Project Topic

OmniCart is an AI-powered, full-stack E-Commerce Recommender System that integrates Machine Learning, Large Language Models (LLMs), Natural Language Processing (NLP), and workflow automation into a structured relational database-driven web platform. The system addresses the core problem of information overload in online retail by intelligently analyzing each customer's purchase history, browsing behavior, and submitted reviews to generate highly personalized product recommendations.

The AI layer of ***OmniCart*** operates across multiple dimensions: a Collaborative Filtering engine (SVD matrix factorization) trained on the Amazon Product Reviews Dataset predicts which products a customer is most likely to purchase next; a cosine similarity search algorithm retrieves the most relevant products from the catalog; a TextBlob/NLTK sentiment analysis pipeline processes review text and scores sentiment; and the OpenAI GPT-4 API generates natural language explanations for every recommendation displayed to the customer. An n8n automation layer ties everything together by triggering real-time and scheduled communications including order confirmations, cart abandonment reminders, and weekly personalized deal digests.

1.2 Objective

The primary objectives of ***OmniCart*** from an AI perspective are:

- Develop a hybrid product recommendation engine combining Collaborative Filtering (SVD) and Content-Based Filtering (TF-IDF + cosine similarity) to deliver accurate, personalized product suggestions for every customer.
- Integrate the OpenAI GPT-4 API to generate human-readable, context-aware explanations for each recommendation and to auto-generate compelling product descriptions for sellers.
- Apply NLP-based sentiment analysis on customer reviews to filter and rank products by genuine community reception, feeding cleaner signals into the recommendation engine.
- Implement a k-Nearest Neighbors (k-NN) search algorithm over product feature vectors to efficiently retrieve the top-N most relevant products from the catalog in real time.
- Automate key customer engagement workflows using n8n, including order confirmations, cart abandonment nudges, weekly recommendation digests, and low-stock seller alerts.

2. System Description

2.1 Original Problem Background

Modern e-commerce platforms host thousands to millions of products across hundreds of categories. Customers routinely experience decision fatigue when browsing generic product listings that do not account for their individual preferences, past purchases, or current intent. Traditional keyword-based search returns results based on exact matches rather than intelligent relevance, leading to poor product discovery and low conversion rates.

On the seller side, writing compelling product descriptions for each listing is time-consuming and often results in inconsistent quality. On the platform side, engaging customers after they leave the platform through timely, relevant communications is typically a manual and reactive process rather than an intelligent, automated one. ***OmniCart*** directly addresses all three of these problems by embedding AI decision-making at every critical touchpoint of the e-commerce workflow.

2.2 Innovations Introduced

- Hybrid Recommendation Engine: ***OmniCart*** combines Collaborative Filtering (SVD matrix factorization via the Surprise library) with Content-Based Filtering (TF-IDF vectorization on product metadata) into a single hybrid pipeline. This overcomes the cold-start problem for new users while maintaining high accuracy for returning customers. The Amazon Product Reviews Dataset (Kaggle, 233M+ reviews) is used to train the SVD model offline, which is then loaded at runtime for real-time predictions.
- LLM-Powered Recommendation Explanations: Rather than showing a bare list of suggested products, ***OmniCart*** uses the OpenAI GPT-4 API to generate a personalized, conversational explanation for each recommendation. For example: 'We think you will love this Sony speaker because you recently purchased wireless earbuds and frequently browse audio accessories.' This transparency builds customer trust and increases click-through rates.
- AI Product Description Generator: When a seller adds a new product listing, they can request an AI-generated description by providing the product name, category, and key features. GPT-4 returns a polished, marketing-quality description that is immediately saved to the database, eliminating the need for manual copywriting.
- NLP Sentiment Analysis Pipeline: Every customer review submitted to ***OmniCart*** is automatically processed by a TextBlob/NLTK sentiment analysis model. A sentiment score between 0.0 (most negative) and 1.0 (most positive) is computed and stored in the Reviews table. The recommendation engine uses these scores as a quality filter, deprioritizing products with consistently negative sentiment.
- k-NN Cosine Similarity Search: Product feature vectors are constructed from TF-IDF representations of product name, category, brand, and description. When a recommendation query is issued, a k-Nearest Neighbors search using

cosine similarity retrieves the top-N most relevant products efficiently. This serves as the Content-Based Filtering component of the hybrid engine.

- Automated n8n Workflow Layer: ***OmniCart*** uses n8n to automate all triggered and scheduled communications. Order confirmations are sent instantly via webhook. Cart abandonment emails fire after 24 hours of inactivity. Weekly personalized deal digests are scheduled every Monday morning. Seller low-stock alerts fire whenever product stock drops below 10 units

3. AI Approach and Methodology

3.1 Search Algorithms

OmniCart employs the following search and retrieval algorithms as part of its AI pipeline:

Algorithm	Application in OmniCart
Cosine Similarity	Measures the similarity between a query product vector and all product feature vectors in the catalog. Used as the core ranking function in the Content-Based Filtering component to identify the most relevant products.
k-Nearest Neighbors (k-NN)	Retrieves the top-N most similar products from the TF-IDF product embedding space. Applied at query time to efficiently find candidate recommendations without exhaustive comparison.
TF-IDF Vectorization	Converts product metadata (name, category, brand, description) into numerical feature vectors. These vectors form the search space over which cosine similarity and k-NN operate.
SVD Matrix Factorization	Decomposes the user-product rating matrix into latent factor matrices. Collaborative Filtering predictions are derived from the dot product of user and product latent vectors.

3.2 LLM / OpenAI API Integration

OmniCart integrates the OpenAI GPT-4 API in two distinct roles within the system:

GPT-4 Use Case	Description
Recommendation Explanation	After the ML engine selects the top-5 products for a customer, each product and the customer's purchase history summary are passed to GPT-4 via a structured prompt. GPT-4 returns a 2-3 sentence personalized explanation that is displayed alongside the recommendation in the UI and stored in the Recommendations table.
Product Description Generator	Sellers provide a product name, category, and bullet-point features. These are passed to GPT-4 with a prompt engineered to produce marketing-quality product descriptions. The generated text is pre-filled into the product listing form and saved to the Products table upon submission.

All prompts are engineered with role-based system messages and structured user messages to ensure consistent, high-quality, and safe output. Temperature is set to 0.7 for recommendation explanations and 0.4 for product descriptions.

3.3 Automation Tools — n8n Workflows

OmniCart uses n8n (self-hosted locally) connected to the Flask backend via HTTP webhooks and scheduled CRON triggers. The following four workflows are implemented:

Workflow	Trigger	Action
Order Confirmation	HTTP Webhook — fires immediately when Flask places an order in DB	Sends formatted order confirmation email to customer with order ID, items, total amount, and estimated delivery.
Cart Abandonment Reminder	CRON — every 24 hours	Queries Cart table for items added more than 24 hours ago with no subsequent order. Sends personalized reminder email to each such customer with cart contents.
Weekly Deals Digest	CRON — every Monday at 9:00 AM	Queries Recommendations table for each customer's top 5 weekly recommendations. Sends a personalized HTML email digest with product names, prices, and AI explanations.

Low Stock Alert	CRON — every 6 hours	Queries Products table for stock_qty < 10. Sends alert email to the relevant seller listing which products are running low and current stock levels.

3.4 Machine Learning Models

The following ML models and techniques are used within ***OmniCart***'s AI pipeline:

Model / Technique	Role in System
SVD Collaborative Filtering (Surprise library)	Core recommendation model. Trained on Amazon Product Reviews Dataset. Predicts a rating score for any user-product pair. Top-N products with highest predicted scores are selected as recommendations.
TF-IDF Vectorizer (Scikit-learn)	Converts product metadata into numerical feature vectors for Content-Based Filtering. Fitted on the full product catalog and used at query time to vectorize searched products.
k-Nearest Neighbors (Scikit-learn)	Retrieves top-N similar products from TF-IDF vector space. Used for content-based and cold-start recommendations where user history is limited.
Sentiment Analysis (TextBlob/NLTK)	Processes raw review text submitted by customers. Outputs a polarity score (0.0 to 1.0). Score is stored in the Reviews table and used as a quality signal in hybrid recommendation ranking.
Hybrid Scoring Function	Combines SVD collaborative score and cosine similarity content score using a weighted average: alpha * CF_score + (1-alpha) * CB_score. Alpha is tuned based on user interaction history volume.

4. System Rules and Mechanics

4.1 System Workflow

The complete ***OmniCart*** AI workflow follows this end-to-end pipeline from user interaction to automated output:

Step	Component	Description
1	Customer Login	Customer logs in and visits homepage. Browsing and search behavior is logged into the database implicitly via the Flask API.
2	Profile Construction	Flask backend queries the customer's Orders, Order_Items, Reviews, and Cart history from MySQL to construct a user interaction profile.
3	Collaborative Filtering	SVD model predicts rating scores for all products not yet purchased. Top-20 candidate products are selected by predicted score.
4	Content-Based Filtering	TF-IDF + k-NN searches the product embedding space using the customer's most-viewed categories as query vector. Returns top-20 content-similar candidates.
5	Sentiment Filtering	Both candidate lists are filtered to remove products with sentiment_score below 0.4 in the Reviews table (consistently negatively reviewed).
6	Hybrid Ranking	Remaining candidates are merged and re-ranked using the weighted hybrid scoring function. Top-5 final recommendations are selected.
7	LLM Explanation	Top-5 products and customer purchase history summary are sent to GPT-4 API. GPT-4 returns a personalized 2-3 sentence explanation for each. Results inserted into Recommendations table.
8	UI Display	Recommendations shown on customer homepage with product details, price, rating, and GPT-4 explanation. Customer can add directly to cart from the recommendations panel.
9	n8n Automation	On order placement, webhook fires order confirmation email. Every Monday, digest email with weekly top-5 recommendations sent automatically to all customers.

4.2 Output / Success Conditions

- The recommendation engine successfully returns top-5 personalized product recommendations for any logged-in customer within 2 seconds of page load.
- Each recommendation is accompanied by a GPT-4 generated explanation that references the customer's actual purchase or browsing history.
- Sentiment analysis correctly classifies at least 85% of reviews as positive, neutral, or negative based on TextBlob polarity scoring.
- The n8n order confirmation workflow fires within 30 seconds of every successful order placement.
- The weekly digest n8n workflow successfully sends personalized emails to all registered customers every Monday at 9:00 AM.
- Recommendation quality is evaluated using Precision@5 and RMSE on a held-out test split of the Amazon Reviews Dataset.

4.3 Process Sequence Summary

Customer logs in → Purchase and review history queried from MySQL → SVD Collaborative Filtering generates candidate list → TF-IDF + k-NN Content-Based Filtering generates second candidate list → Sentiment filter removes low-quality products → Hybrid scoring merges and ranks candidates → Top-5 sent to GPT-4 API for explanation generation → Results inserted into Recommendations table in MySQL → Displayed on customer UI with personalized explanations → n8n automation handles all downstream communications automatically.

5. Libraries and Tools

Category	Tool / Library	Purpose
LLM API	OpenAI API (GPT-4)	Recommendation explanation generation and AI product description writing for sellers
Automation	n8n (self-hosted)	Order confirmation, cart abandonment, weekly digest, and low-stock alert workflows
ML — Recommender	Surprise (scikit-surprise)	SVD matrix factorization for Collaborative Filtering trained on Amazon Reviews Dataset
ML — Search	Scikit-learn	TF-IDF vectorization, cosine similarity computation, k-NN product search
NLP — Sentiment	TextBlob / NLTK	Sentiment polarity analysis on customer review text; scores stored in Reviews table
Backend Framework	Python 3.11 + Flask	REST API serving all frontend requests and triggering the AI recommendation pipeline
Database ORM	SQLAlchemy	MySQL database interaction, model definition, and query execution from Flask
Authentication	Flask-JWT-Extended	JWT token-based authentication and role-based access control (Customer/Seller/Admin)
Data Processing	Pandas + NumPy	Dataset loading, preprocessing, and matrix operations for offline model training
Frontend	React.js + Tailwind CSS	Dynamic, responsive user interface for all three role-based portals
Analytics Charts	Chart.js	Visual analytics graphs on admin and seller dashboards
Database	MySQL 8.0	Primary relational database storing all 9 tables including the Recommendations table
Dataset	Amazon Product Reviews (Kaggle)	Training dataset for SVD Collaborative Filtering model (233 million+ reviews across 29 categories)

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